IonVe
ELECTROCHEMICAL REACTORS

Long-Duration Industrial Energy Storage

IonVe develops redox flow battery systems targeting 4–12 hour storage for heavy industry. Architecture redesigned from cell level. Design-target phase.

TRL 4-5 · Pre-incorporation · June 2026

Why long-duration storage, and why now?

Industrial demand charges are the real cost driver

European industries face rising, volatile electricity costs as industrial processes become increasingly electrified. For energy-intensive sectors — steel, foundries, cement and recycling — electricity directly affects margins and competitiveness. For them, **energy management is a competitiveness factor**.

Lithium-ion is not cost-effective beyond 4–6 hours

Current energy storage systems, especially lithium-ion batteries, are not always suitable for these applications. Fire risk, degradation (replacements required) and short duration (technical limitations) **make the investment unattractive**.

Existing flow batteries are still too expensive

Flow batteries are naturally suited for long-duration industrial storage. They are **safe, non-flammable, stable over long lifetimes**. However, current systems remain expensive. High component count, complex stack assembly and site-specific engineering make stall their implementation.

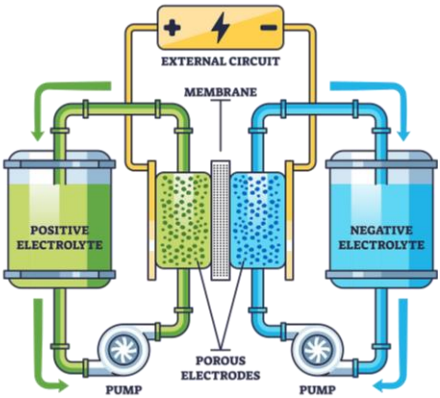
What is a flow battery?

Think of a liquid-filled tank that produces electricity as the fluid passes through an electrochemical cell.

More tank volume = more hours of storage

More stacked cells = more power

This makes flow batteries well suited for large-scale and long-duration energy storage.



Lithium-ion

- Fire-safety constraints
- Degrades with daily cycling
- Replacement/upgrade often needed
- Best for short duration <4 h

Flow Battery

- Non-flammable electrolyte
- 30+ yr lifespan, no capacity loss
- 20–30+ year lifetime potential
- Designed for large scale & long duration

IonVe — We Introduce Three Architecture Innovations

That make a more efficient, cheap and scalable flow battery system for industries

297	>30	~5
€/kWh (6h duration)	Years – operating life	Years – Payback Time

INNOVATION 01

Conductive Polymer Frame

A new multifunctional component, based to a new patented material that replace graphite, improve manufacturability, reduce part count and enable scalable assembly.

Estimated cost reduction: 30–40% per cell

INNOVATION 02

High-Power Ministack Module

A standardized power unit with +20% power output and +7% efficiency. IonVe system uses 40% less modules of a conventional RFB, dramatically reducing costs.

8x less volume for same power & +5 pp RTE

INNOVATION 03

Modular Architecture

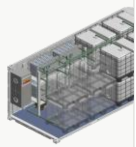
A modular architecture that improves scalability, control, reconfigurability and maintenance, simplifying power electronics and improving round-trip efficiency (RTE).

Assembly cost: -20%

THE FINAL PRODUCT

Integrated System Platform

The three innovations come together in one control-driven product platform, built for efficiency, repeatability and series production.



Power range: 100kW to 500MW — same architecture

-40%	+10%	+20%
Unit Capital Cost	Round Trip Efficiency	Operational Range

Modelled annual savings of a reference mid-size European foundry implementing IonVe system

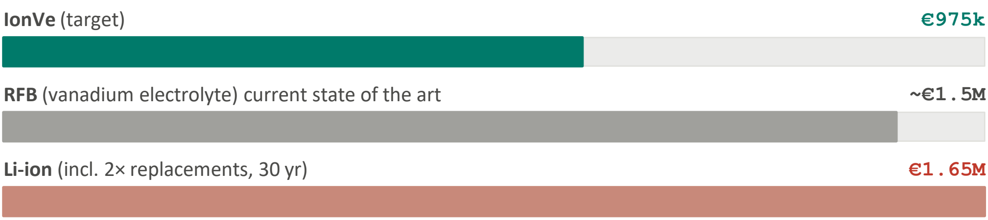
+€3.5m / yr
avg. electricity bill
Mid size European Foundry
500 kW / 3 MWh reference RFB

ACTUAL SCENARIO (TODAY)
-10%
on electricity bill
€350k saved / year

FUTURE SCENARIO (2035)
-25%
on electricity bill
+€875k saved / year

Savings model: ~65% from demand-charge reduction (peak shaving), ~35% from time-of-use arbitrage.

30-year TCO comparison — 500 kW / 3 MWh



ROADMAP & NEXT STEPS

1. Proof of Concept – Architecture Freeze · Jan–Jun 2026 · TRL 3→5 ← **We are here**

MK1/2 hydraulic & electrical tests · 3 go/no-go criteria · Patent filings (3 applications)

2. Industrial Pilot · H2 2026 – H1 2027 · TRL 5→7

2–3 co-development pilots · Supply-chain qualification

3. Commercial Demonstration · 2027-2028 · TRL 7→8

First commercial demonstration & contracts · Strategic round for scale-up

IonVe MK1 prototype ministack modules in test

4. Market Scale · 2028 + · TRL 8→9

15–30 systems/yr, 5 countries · EPC channel partnerships · Energy-as-a-Service model

How we work with partners

01 Site-Specific Screening — No cost, no obligation

We model annual savings from industries 15-min interval load data and electricity contract. Output: site report with NPV, payback, and sensitivity within 2 weeks of data receipt.

02 Access to Test Data — Under NDA, on request

We share electrochemical results and stack performance data as milestones are completed.

03 Co-Development Pilot — Post-PoC, from Q3 2026

2–3 co-developed pilot deployments with industrial partners, structured on cost-plus terms to generate field operating data, integration feedback and independent performance validation.

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Visit our website for more information!

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All performance and cost figures are design targets. Validation in progress.